Topic: Power series solutions

Question: Find a power series solution to the differential equation.

$$y' - y = 0$$

Answer choices:

A $y = e^x$

B $y = c_0 e^x$

C $y = c_0 x e^x$

D $y = c_0 x e^{x^2}$

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Solution: B

We'll use the standard form for the power series solution and its derivative,

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

to substitute into the differential equation.

$$y' - y = 0$$

$$\sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

The first series will start with the $x^{1-1} = x^0$ term, while the second series will start with the x^0 term, so the series are already in phase. To make the indices match, we'll substitute $k = n - 1$ and $n = k + 1$ into the first series (the new index will be $k = 1 - 1 = 0$), and $k = n$ into the second series (the new index will be $k = 0$).

$$\sum_{k=0}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=0}^{\infty} c_k x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$\sum_{k=0}^{\infty} c_{k+1}(k+1)x^k - c_k x^k = 0$$

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$$\sum_{k=0}^{\infty}(c_{k+1}(k+1)-c_k)x^k=0$$

Equating coefficients and then solving for the coefficient with the largest subscript, c_{k+1}, gives the recurrence relation.

$$c_{k+1}(k+1)-c_k=0$$

$$c_{k+1}=\frac{c_k}{k+1}$$

Plugging in $k=0,\ 1,\ 2,\ 3,\ \ldots$, we start to see a pattern.

$$k=0 \qquad c_{0+1}=\frac{c_0}{0+1} \qquad c_1=\frac{c_0}{1!}$$

$$k=1 \qquad c_{1+1}=\frac{c_1}{1+1} \qquad c_2=\frac{c_0}{2!}$$

$$k=2 \qquad c_{2+1}=\frac{c_2}{2+1} \qquad c_3=\frac{c_0}{3!}$$

$$k=3 \qquad c_{3+1}=\frac{c_3}{3+1} \qquad c_4=\frac{c_0}{4!}$$

Plugging these coefficients into the expansion of the power series solution, we get

$$y=c_0+c_1x+c_2x^2+c_3x^3+c_4x^4+\ldots$$

$$y=c_0+\frac{c_0}{1!}x+\frac{c_0}{2!}x^2+\frac{c_0}{3!}x^3+\frac{c_0}{4!}x^4+\ldots$$

$$y=c_0\left(\frac{1}{0!}x^0+\frac{1}{1!}x^1+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\frac{1}{4!}x^4+\ldots\right)$$

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We can rewrite the series inside the parentheses.

$$y = c_0 \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

We recognize this series as the common Maclaurin series representation of e^x,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

so the general solution to the differential equation is

$$y = c_0 e^x$$

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Topic: Power series solutions

Question: Find a power series solution to the differential equation.

$$y' = x^2 y$$

Answer choices:

A $y(x) = c_0 e^{x}$

B $y(x) = c_0 e^{\frac{x^2}{2}}$

C $y(x) = c_0 e^{\frac{x^3}{3}}$

D $y(x) = c_0 e^{\frac{x^4}{4}}$

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Solution: C

We'll use the standard form for the power series solution and its derivative,

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

to substitute into the differential equation.

$$y' = x^2 y$$

$$y' - x^2 y = 0$$

$$\sum_{n=1}^{\infty} c_n n x^{n-1} - x^2 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+2} = 0$$

The first series will start with the $x^{1-1} = x^0$ term, while the second series will start with the $x^{0+2} = x^2$ term, so we'll pull out the x^0 and x^1 terms from the first series in order to put the series in phase.

$$c_1(1)x^{1-1} + c_2(2)x^{2-1} + \sum_{n=3}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+2} = 0$$

$$c_1 x^0 + 2c_2 x^1 + \sum_{n=3}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+2} = 0$$

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$$c_1 + 2c_2x + \sum_{n=3}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^{n+2} = 0$$

Now the series are in phase (they both begin with the x^2 term), so we just need to make their indices match. We'll substitute $k = n - 1$ and $n = k + 1$ into the first series (the new index will be $k = 3 - 1 = 2$), and $k = n + 2$ and $n = k - 2$ into the second series (the new index will be $k = 2$).

$$c_1 + 2c_2x + \sum_{k=2}^{\infty} c_{k+1}(k+1)x^k - \sum_{k=2}^{\infty} c_{k-2}x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$c_1 + 2c_2x + \sum_{k=2}^{\infty} c_{k+1}(k+1)x^k - c_{k-2}x^k = 0$$

$$c_1 + 2c_2x + \sum_{k=2}^{\infty} (c_{k+1}(k+1) - c_{k-2})x^k = 0$$

Equating coefficients, we get

$$c_1 = 0 \qquad k = 0$$

$$2c_2 = 0 \qquad k = 1$$

$$c_{k+1}(k+1) - c_{k-2} = 0 \qquad k = 2, 3, 4, 5, \ldots$$

Solving the recurrence relation for the coefficient with the largest subscript, c_{k+1}, gives

$$c_1 = 0 \qquad k = 0$$

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$$c_2 = 0 \qquad k = 1$$

$$c_{k+1} = \frac{c_{k-2}}{k+1} \qquad k = 2, 3, 4, 5, \ldots$$

Plugging in $k = 2, 3, 4, 5, \ldots$, we start to see a pattern.

$$k = 0 \quad c_1 = 0 \qquad k = 1 \quad c_2 = 0$$

$$k = 2 \quad c_3 = \frac{c_0}{3} \qquad k = 3 \quad c_4 = 0 \qquad k = 4 \quad c_5 = 0$$

$$k = 5 \quad c_6 = \frac{c_0}{3 \cdot 6} \qquad k = 6 \quad c_7 = 0 \qquad k = 7 \quad c_8 = 0$$

$$k = 8 \quad c_9 = \frac{c_0}{3 \cdot 6 \cdot 9} \qquad k = 9 \quad c_{10} = 0 \qquad k = 10 \quad c_{11} = 0$$

$$k = 11 \quad c_{12} = \frac{c_0}{3 \cdot 6 \cdot 9 \cdot 12} \qquad \ldots \qquad \ldots$$

...

Plugging these coefficients into the expansion of the power series solution, we get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \ldots$$

$$y = c_0 + (0)x + (0)x^2 + \frac{c_0}{3}x^3 + (0)x^4 + (0)x^5 + \frac{c_0}{3 \cdot 6}x^6$$

$$+(0)x^7 + (0)x^8 + \frac{c_0}{3 \cdot 6 \cdot 9}x^9 + \ldots$$

$$y = c_0 + \frac{c_0}{3}x^3 + \frac{c_0}{3 \cdot 6}x^6 + \frac{c_0}{3 \cdot 6 \cdot 9}x^9 + \frac{c_0}{3 \cdot 6 \cdot 9 \cdot 12}x^{12} + \ldots$$

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$$y = c_0\left(1 + \frac{1}{3}x^3 + \frac{1}{3\cdot 6}x^6 + \frac{1}{3\cdot 6\cdot 9}x^9 + \frac{1}{3\cdot 6\cdot 9\cdot 12}x^{12} + \ldots\right)$$

$$y = c_0\left(1 + \frac{1}{3^1}x^3 + \frac{1}{3^2(1\cdot 2)}x^6 + \frac{1}{3^3(1\cdot 2\cdot 3)}x^9 + \frac{1}{3^4(1\cdot 2\cdot 3\cdot 4)}x^{12} + \ldots\right)$$

$$y = c_0\left(\frac{1}{3^0(0!)}x^0 + \frac{1}{3^1(1!)}x^{3(1)} + \frac{1}{3^2(2!)}x^{3(2)} + \frac{1}{3^3(3!)}x^{3(3)} + \frac{1}{3^4(4!)}x^{3(4)} + \ldots\right)$$

We can rewrite the series inside the parentheses.

$$y = c_0\sum_{n=0}^{\infty}\frac{1}{3^n n!}x^{3n}$$

We could leave the series solution this way, but we can also rewrite the series as

$$y = c_0\sum_{n=0}^{\infty}\frac{(x^3)^n}{3^n n!}$$

$$y = c_0\sum_{n=0}^{\infty}\left(\frac{x^3}{3}\right)^n\frac{1}{n!}$$

$$y = c_0\sum_{n=0}^{\infty}\frac{\left(\frac{x^3}{3}\right)^n}{n!}$$

We recognize this series as the common Maclaurin series representation of e^x, except with $x^3/3$ instead of just x.

$$e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!}$$

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$$e^{\frac{x^3}{3}} = \sum_{n=0}^{\infty} \frac{\left(\frac{x^3}{3}\right)^n}{n!}$$

So the general solution to the differential equation is

$$y = c_0 e^{\frac{x^3}{3}}$$

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Topic: Power series solutions

Question: Find a power series solution to the differential equation.

$$y'' + xy' + y = 0$$

Answer choices:

A $$y(x) = c_0 \sum_{n=0}^{\infty} \frac{(1)^n}{2^n n!} x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{(2)^n n!}{(2n+1)!} x^{2n+1}$$

B $$y(x) = c_0 \sum_{n=0}^{\infty} \frac{(1)^n}{2^n n!} x^{2n} + c_1 \sum_{n=1}^{\infty} \frac{(2)^n n!}{(2n+1)!} x^{2n+1}$$

C $$y(x) = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^{2n} + c_1 \sum_{n=1}^{\infty} \frac{(-2)^n n!}{(2n+1)!} x^{2n+1}$$

D $$y(x) = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{(-2)^n n!}{(2n+1)!} x^{2n+1}$$

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Solution: D

We'll use the standard form for the power series solution and its derivatives,

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

to substitute into the differential equation.

$$y'' + xy' + y = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} + x \sum_{n=1}^{\infty} c_n n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} + \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

The first series will start with the $x^{2-2} = x^0$ term, the second series will start with the x^1 term, and the third series will start with the x^0 term. So we'll pull out the x^0 term from both the first and the third series in order to put the series in phase.

$$c_2(2)(2-1)x^{2-2} + \sum_{n=3}^{\infty} c_n n(n-1) x^{n-2} + \sum_{n=1}^{\infty} c_n n x^n + c_0 x^0 + \sum_{n=1}^{\infty} c_n x^n = 0$$

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$$c_0 + 2c_2 + \sum_{n=3}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=1}^{\infty} c_n x^n = 0$$

Now the series are in phase (they all begin with the x^1 term), so we just need to make their indices match. We'll substitute $k = n - 2$ and $n = k + 2$ into the first series (the new index will be $k = 3 - 2 = 1$), and we'll substitute $k = n$ into the second and third series (their new indices will be $k = 1$).

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+1)x^k + \sum_{k=1}^{\infty} c_k k x^k + \sum_{k=1}^{\infty} c_k x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+1)x^k + c_k k x^k + c_k x^k = 0$$

$$c_0 + 2c_2 + \sum_{k=1}^{\infty} (c_{k+2}(k+2)(k+1) + c_k k + c_k)x^k = 0$$

Equating coefficients, we get

$$c_0 + 2c_2 = 0 \qquad k = 0$$

$$c_{k+2}(k+2)(k+1) + c_k k + c_k = 0 \qquad k = 1, 2, 3, 4, \ldots$$

Solving the recurrence relation for the coefficient with the largest subscript, c_{k+2}, gives

$$c_0 + 2c_2 = 0 \qquad k = 0$$

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$$c_{k+2} = -\frac{c_k}{k+2} \qquad k = 1, 2, 3, 4, \ldots$$

Plugging in $k = 1, 2, 3, 4, \ldots$, we start to see a pattern.

$$k = 0 \qquad c_2 = -\frac{c_0}{2} \qquad k = 1 \qquad c_3 = -\frac{c_1}{3}$$

$$k = 2 \qquad c_4 = \frac{c_0}{2 \cdot 4} \qquad k = 3 \qquad c_5 = \frac{c_1}{3 \cdot 5}$$

$$k = 4 \qquad c_6 = -\frac{c_0}{2 \cdot 4 \cdot 6} \qquad k = 5 \qquad c_7 = -\frac{c_1}{3 \cdot 5 \cdot 7}$$

... ...

Plugging these coefficients into the expansion of the power series solution, we get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \ldots$$

$$y = c_0 + c_1 x - \frac{c_0}{2} x^2 - \frac{c_1}{3} x^3 + \frac{c_0}{2 \cdot 4} x^4 + \frac{c_1}{3 \cdot 5} x^5 - \frac{c_0}{2 \cdot 4 \cdot 6} x^6 - \frac{c_1}{3 \cdot 5 \cdot 7} x^7 + \ldots$$

$$y = \left(c_0 - \frac{c_0}{2} x^2 + \frac{c_0}{2 \cdot 4} x^4 - \frac{c_0}{2 \cdot 4 \cdot 6} x^6 + \ldots \right)$$

$$+ \left(c_1 x - \frac{c_1}{3} x^3 + \frac{c_1}{3 \cdot 5} x^5 - \frac{c_1}{3 \cdot 5 \cdot 7} x^7 + \ldots \right)$$

$$y = c_0 \left(1 - \frac{1}{2} x^2 + \frac{1}{2 \cdot 4} x^4 - \frac{1}{2 \cdot 4 \cdot 6} x^6 + \ldots \right)$$

$$+ c_1 \left(x - \frac{1}{3} x^3 + \frac{1}{3 \cdot 5} x^5 - \frac{1}{3 \cdot 5 \cdot 7} x^7 + \ldots \right)$$

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We could leave the solution this way, or we could try to rewrite each series in order to simplify the general solution.

$$y = c_0\left(\frac{1}{2^0(0!)}x^{2(0)} - \frac{1}{2^1(1!)}x^{2(1)} + \frac{1}{2^2(2!)}x^{2(2)} - \frac{1}{2^3(3!)}x^{2(3)} + \dots\right)$$

$$+c_1\left(x^{2(0)+1} - \frac{1}{3}x^{2(1)+1} + \frac{1}{3\cdot 5}x^{2(2)+1} - \frac{1}{3\cdot 5\cdot 7}x^{2(3)+1} + \dots\right)$$

$$y = c_0\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n n!}x^{2n}$$

$$+c_1\left(x^{2(0)+1} - \frac{1}{3}x^{2(1)+1} + \frac{1}{3\cdot 5}x^{2(2)+1} - \frac{1}{3\cdot 5\cdot 7}x^{2(3)+1} + \dots\right)$$

For the c_1 series, we want to try representing the denominator with factorial notation, but the denominators don't include even numbers. So we'll have to add those into the numerator to cancel them out when we switch to factorial notation.

$$y = c_0\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n n!}x^{2n}$$

$$+c_1\left(x^{2(0)+1} - \frac{1(2)}{(2(1)+1)!}x^{2(1)+1} + \frac{1(2\cdot 4)}{(2(2)+1)!}x^{2(2)+1} - \frac{1(2\cdot 4\cdot 6)}{(2(3)+1)!}x^{2(3)+1} + \dots\right)$$

$$y = c_0\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n n!}x^{2n}$$

$$+c_1\left(x^{2(0)+1} - \frac{1(2^1)}{(2(1)+1)!}x^{2(1)+1} + \frac{1(2^2)(1\cdot 2)}{(2(2)+1)!}x^{2(2)+1} - \frac{1(2^3)(1\cdot 2\cdot 3)}{(2(3)+1)!}x^{2(3)+1} + \dots\right)$$

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$$y = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^{2n}$$

$$+c_1 \left(\frac{2^0(0!)}{(2(0)+1)!} x^{2(0)+1} - \frac{2^1(1!)}{(2(1)+1)!} x^{2(1)+1} + \frac{2^2(2!)}{(2(2)+1)!} x^{2(2)+1} - \frac{2^3(3!)}{(2(3)+1)!} x^{2(3)+1} + \ldots \right)$$

So the general solution to the differential equation is

$$y = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{(-2)^n n!}{(2n+1)!} x^{2n+1}$$

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